A PODI-Based Reduced-Order Model for Air-Quality Prediction

Reduced-order modelling of pollutant dispersion across mobility scenarios

Proper Orthogonal Decomposition with Interpolation

Proper Orthogonal Decomposition with Interpolation (PODI) is a reduced-order modelling technique used to approximate high-dimensional simulation fields with a small number of dominant spatial patterns. The main idea is to learn these patterns from existing CFD snapshots, represent each full concentration field through a compact set of modal coefficients, and then interpolate those coefficients for new operating conditions.

In this work, the already available dispersion data are used to build a reduced-order model for pollutant concentration prediction. Instead of running a new full CFD simulation for every hour, wind condition, or emission scenario, the PODI model reconstructs the concentration field from the learned modes. This makes it possible to evaluate full-day pollutant behaviour and compare mobility scenarios at a much lower computational cost, while still retaining the dominant spatial structure contained in the original simulations. The objective is therefore to turn the available CFD data into a practical reduced-order model that can support rapid scenario screening and receptor-level exposure assessment.

1 Input data and labelling

Snapshots. The training data consist of converged pollutant concentration fields obtained for a reference condition and several emission-reduction scenarios. Each snapshot represents the spatial concentration field for one pollutant at one hour. Representative hours are selected to cover variations in wind speed, wind direction, and emission strength.

Labels. Each snapshot is associated with a compact physical parameter vector, $p = (u, v, G, L)$. Here, (u, v) are the two horizontal wind components, G is a global emission scaling factor, and L is a local emission scaling factor applied to a targeted transport corridor. This parameterisation makes it possible to incoprporate interpolation when pollution is reduced in certain tagged roads.

Scenario	Definition	G (global)	L (local)
Reference	Baseline traffic-emission condition	1.0	1.0
S1	Moderate fleet electrification	0.8	1.0
S2	Strong fleet electrification	0.6	1.0
S3	Local corridor emission reduction	1.0	0.5

Table 1: Mobility scenarios expressed as general emission-scale labels for the surrogate.

Training and testing. The available snapshots are split into training and test sets while preserving all scenario types in both sets. The same split is used for each pollutant so that the surrogate accuracy can be compared consistently.

2 Method and Approximation function

Magnitude–shape decoupling. Concentration magnitude can vary strongly because pollutant levels depend on both wind dilution and emission strength. To avoid forcing a single linear model to learn both field magnitude and spatial pattern simultaneously, each snapshot x_c is split into a scalar volume-weighted magnitude and a unit-norm shape,

$$s_c = \|x_c\|_V = \sqrt{x_c^\top V x_c}, \qquad \hat{x}_c = x_c/s_c, \qquad V = \mathrm{diag}(\text{cell volumes}). \tag{1}$$

POD by the method of snapshots. POD is then applied to the normalized shapes. The method of snapshots forms a small Gram matrix from the available fields and extracts the dominant modes without explicitly storing a full high-dimensional basis. This makes the approach suitable even when the original CFD fields contain many millions of cells.

Ansatz. For a new parameter vector $p = (u, v, G, L)$, the pollutant field is reconstructed as

$$\boxed{x(p) \;=\; m(p)\left[\bar{x} \;+\; \sum_{k=1}^{r} \phi_k \, a_k(p)\right]} \tag{2}$$

where $m(p)$ is the predicted magnitude, $\bar{x}$ is the mean normalized field, ϕ_k are the POD modes, and $a_k(p)$ are the interpolated modal coefficients. The coefficient model uses wind and emission parameters, together with simple physics-informed features such as wind-speed magnitude and emission-to-wind scaling.

The magnitude model is written in multiplicative form so that changes in emission strength are handled consistently with passive-scalar physics. For a frozen flow field, pollutant concentration is linear in the emission source strength. Encoding this relation directly helps the surrogate remain physically meaningful instead of acting only as a black-box interpolation tool.

3 Validation study

The number of POD modes is selected by comparing training and test reconstruction errors over a range of retained modes. The error decreases rapidly at first and then reaches a plateau, indicating that the dominant spatial structures have already been captured. Five modes are retained as a compact and stable choice. Table 2 summarises the validation accuracy.

Pollutant	Modes r	Test relative-L_2	Parity R^2
CO	5	0.332	0.963
NO_x	5	0.325	0.963

Table 2: Surrogate accuracy on the held-out test set.

Physical consistency check. In addition to statistical validation, the surrogate is checked against directly simulated data at receptor level. Agreement at the receptor surfaces confirms that the reduced-order reconstruction preserves the quantities used for the final assessment, not only the global field norm.

4 Prediction results

The trained surrogate is evaluated over the complete daily sequence of wind conditions for all mobility scenarios and both pollutants. The reconstructed concentration fields are then sampled

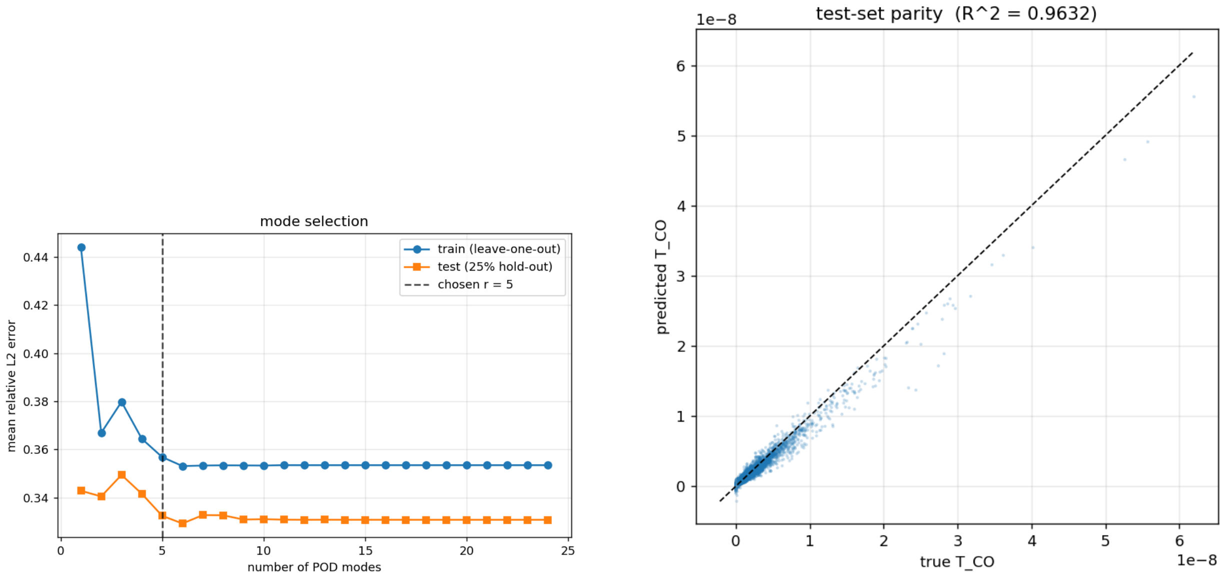

Figure 1: Left: mode-selection curve showing the reconstruction error as the number of retained POD modes increases. Right: parity comparison between predicted and reference cell values for the test set.

at sensitive receptor locations using the same post-processing definition as in the original CFD workflow. This provides a full daily concentration profile at each receptor, together with daily mean and peak values.

CO — daily-mean concentration ($\mu g\,m^{-3}$), change vs Reference					
Receptor	Reference	S1	S2	S3	24h peak (Ref)
Public Hospital	3.68	2.94 (−20.1%)	2.20 (−40.2%)	2.58 (−30.0%)	9.54
Francisco de Holanda	2.04	1.63 (−20.0%)	1.22 (−40.0%)	1.71 (−15.9%)	5.28
Martins Sarmento	3.60	2.88 (−20.0%)	2.16 (−39.9%)	3.04 (−15.4%)	9.56
Santos Simões	4.34	3.47 (−19.9%)	2.61 (−39.9%)	3.86 (−11.0%)	11.66

Table 3: CO daily statistics over the full 24-hour cycle predicted by the surrogate.

$\mathbf{NO_x}$ — daily-mean concentration ($\mu g\,m^{-3}$), change vs Reference					
Receptor	Reference	S1	S2	S3	24h peak (Ref)
Public Hospital	4.98	3.98 (−20.0%)	2.99 (−40.0%)	3.96 (−20.4%)	12.15
Francisco de Holanda	2.87	2.30 (−20.0%)	1.72 (−40.0%)	2.38 (−17.0%)	6.80
Martins Sarmento	5.35	4.28 (−20.0%)	3.21 (−40.0%)	4.45 (−16.9%)	13.00
Santos Simões	6.39	5.12 (−20.0%)	3.84 (−40.0%)	5.47 (−14.5%)	15.47

Table 4: NO_x daily statistics over the full 24-hour cycle predicted by the surrogate.

Findings. The electrification scenarios reduce concentration nearly uniformly across all receptors because they act as global emission reductions. The local corridor intervention is more spatially selective: it provides the strongest improvement near the affected corridor but smaller benefits at receptors farther away. Peak concentrations remain substantially higher than daily mean values, which shows why full-day profiles are more informative than a small number of sampled hours.

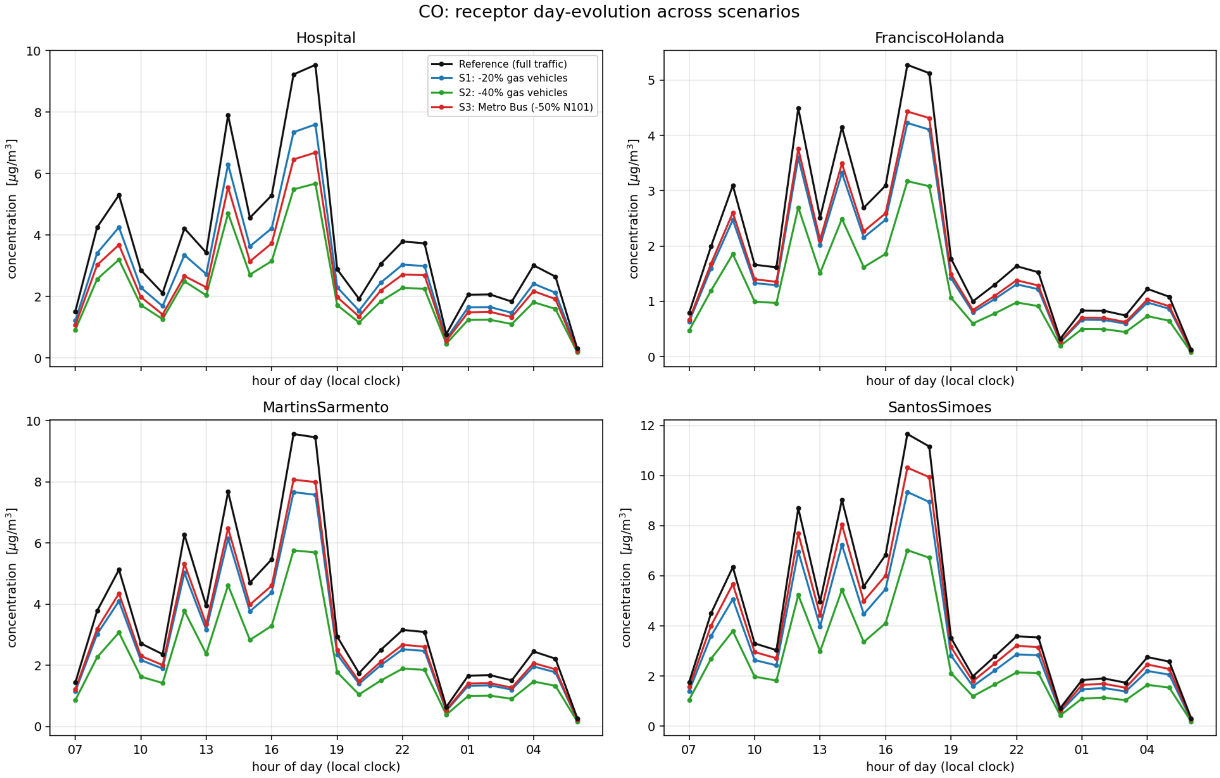

Figure 2: CO daily evolution at the four receptors. Electrification lowers concentration uniformly, whereas the local corridor intervention produces a more spatially selective benefit.

5 Conclusion

The PODI model demonstrates how an existing CFD dataset can be transformed into a fast and useful reduced-order model for air-quality assessment. By learning the dominant spatial patterns from the available concentration fields and interpolating their reduced coefficients, the surrogate can predict full-field pollutant distributions for new wind and emission conditions at a fraction of the cost of direct simulation.

The main value of the approach is that it extends the usefulness of an already completed CFD campaign. Instead of treating the available simulations as isolated cases, PODI connects them into a continuous prediction framework. This enables rapid evaluation of the complete daily cycle, receptor-level exposure, and alternative emission-reduction strategies without repeating the full CFD workflow for every new query.

The results also show that different mobility measures produce different spatial benefits. Global emission reductions give broad and uniform concentration decreases, while local interventions are most effective close to the targeted corridor. Therefore, the most robust air-quality improvement is likely obtained by combining broad fleet-level reductions with targeted local transport measures.

Overall, PODI provides a practical bridge between high-fidelity CFD and decision-oriented scenario analysis. It preserves the key physical information contained in the original simulations while making the assessment fast enough for screening, comparison, and future optimisation studies.

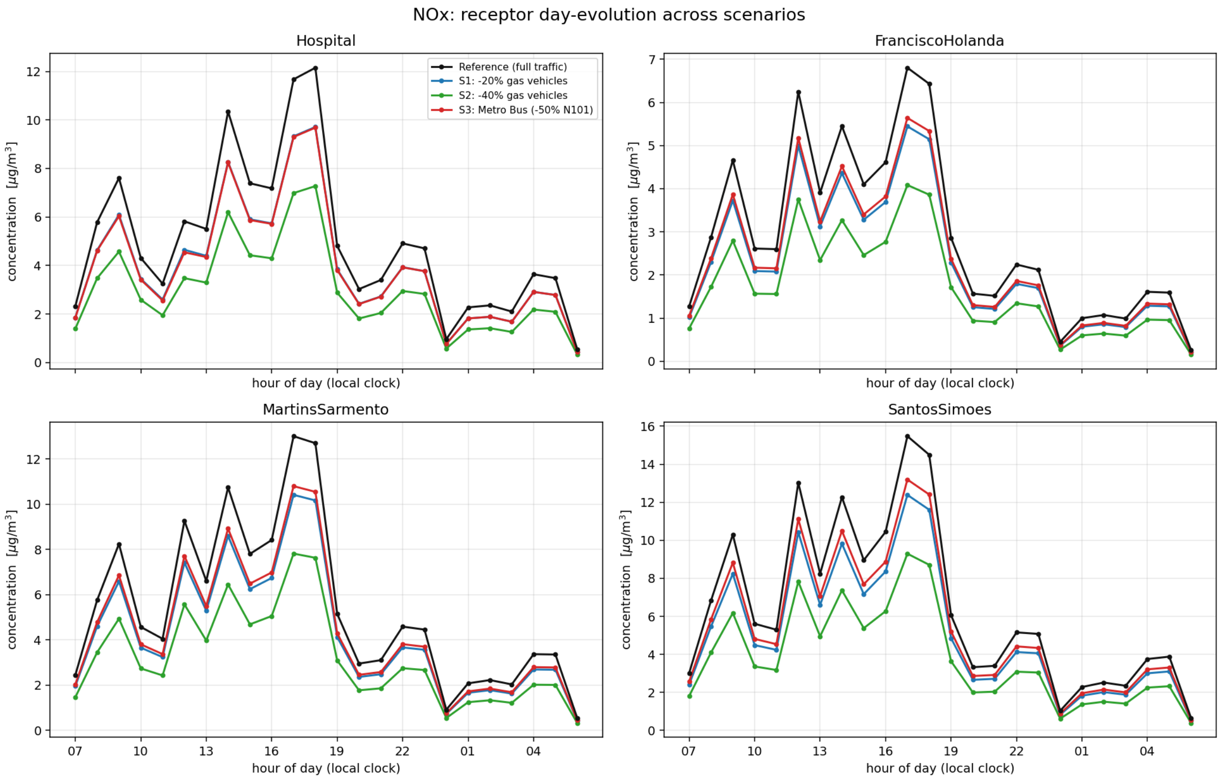

Figure 3: NO_x daily evolution at the four receptors. The temporal trend follows the hourly wind cycle, with higher concentrations during less favourable dispersion periods.